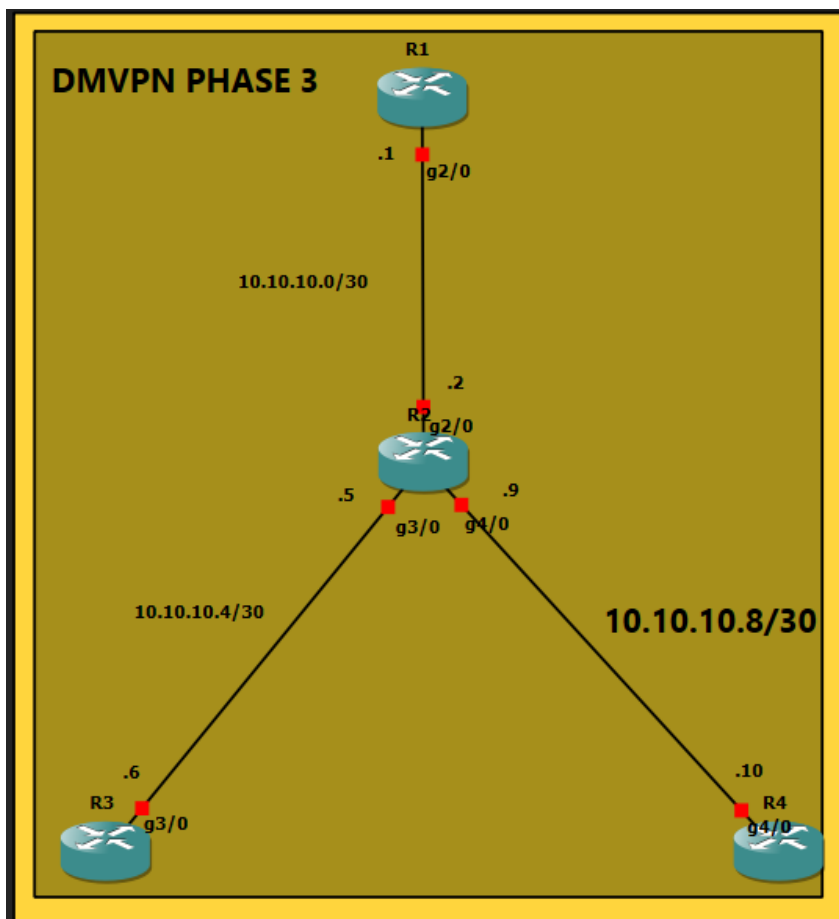


DMVPN PHASE 3 GNS3 PROJECT

Configuration of DMVPN Phase 3 on Cisco routers on GNS3, and comparison of phase 2 and 1.



ISAKMP

To begin with, IKEv1 (the classic ISAKMP configuration) or IKEv2 must be configured, in this lab would be ISAKMP, this configuration must match in every router where dmvpn is being implemented.

DMVPN PHASE 3 GNS3 PROJECT

HUB R1:

```
!  
crypto isakmp policy 1  
  encr aes 256  
  hash sha256  
  authentication pre-share  
  group 2  
crypto isakmp key abc123 address 0.0.0.0
```

SPOKE R3:

```
!  
crypto isakmp policy 1  
  encr aes 256  
  hash sha256  
  authentication pre-share  
  group 2  
crypto isakmp key abc123 address 0.0.0.0
```

SPOKE R4:

```
!  
crypto isakmp policy 1  
  encr aes 256  
  hash sha256  
  authentication pre-share  
  group 2  
crypto isakmp key abc123 address 0.0.0.0
```

IPSEC

IPSEC must also match configuration in every router where DMVPN is implemented.

Hub R1:

```
!  
crypto ipsec transform-set T-SET ah-sha256-hmac esp-aes 256  
  mode tunnel  
!  
crypto ipsec profile IPSEC_PROF  
  set transform-set T-SET  
!
```

DMVPN PHASE 3 GNS3 PROJECT

TUNNEL INTERFACE CONF

The difference is here, when you configure the interface Tunnel on the router HUB the configuration of NHRP is crucial, there are not big differences but is still something to consider making nhrp and dmvpn work.

HUB R1:

```
!  
interface Tunnel10  
 ip address 192.168.1.1 255.255.255.0  
 no ip redirects  
 ip nhrp map multicast dynamic  
 ip nhrp network-id 100  
 ip nhrp redirect  
 tunnel source GigabitEthernet2/0  
 tunnel mode gre multipoint  
 tunnel key 100  
 tunnel protection ipsec profile IPSEC_PROF  
!
```

SPOKE R3:

```
!  
interface Tunnel10  
 ip address 192.168.1.2 255.255.255.0  
 no ip redirects  
 ip nhrp network-id 100  
 ip nhrp nhs 192.168.1.1 nbma 10.10.10.1 multicast  
 ip nhrp shortcut  
 tunnel source GigabitEthernet3/0  
 tunnel mode gre multipoint  
 tunnel key 100  
 tunnel protection ipsec profile IPSEC_PROF  
!
```

SPOKE R4:

```
!  
interface Tunnel10  
 ip address 192.168.1.3 255.255.255.0  
 no ip redirects  
 ip nhrp network-id 100  
 ip nhrp nhs 192.168.1.1 nbma 10.10.10.1 multicast  
 ip nhrp shortcut  
 tunnel source GigabitEthernet4/0  
 tunnel mode gre multipoint  
 tunnel key 100  
 tunnel protection ipsec profile IPSEC_PROF  
end
```